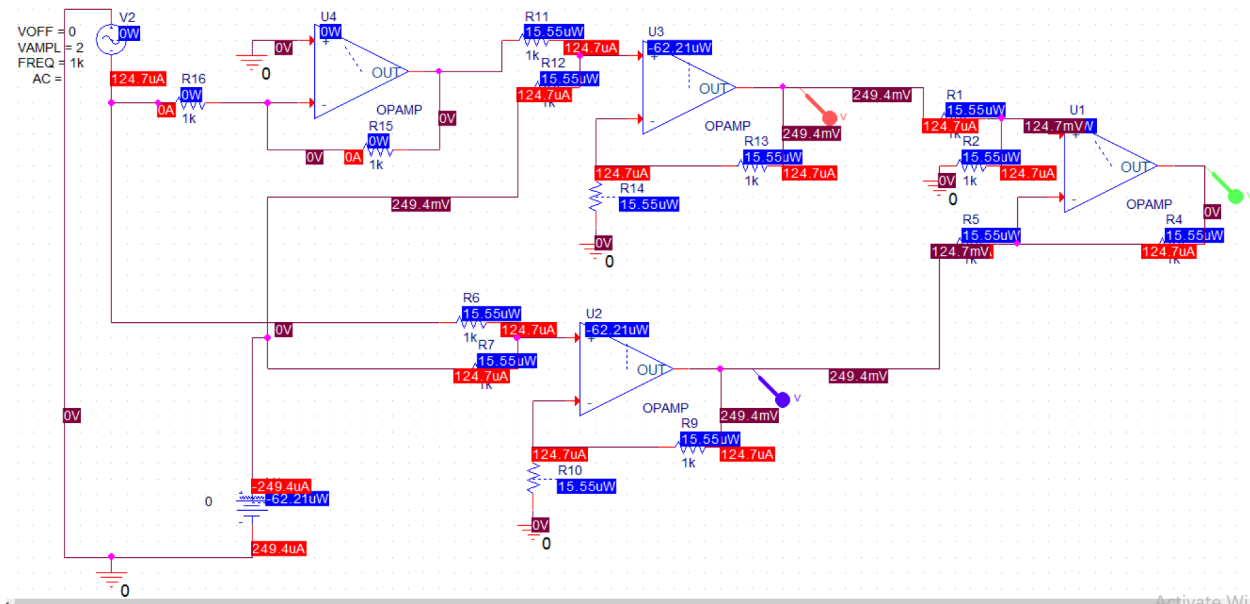


ETE 4521L Data Communication and Networking
Lab 1 Differential Signaling
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OBJECTIVES

- Learn how differential signaling can be used to remove common-mode noise from a signal
- Use Orcad PSpice to design and simulate a circuit to add and remove noise from a signal

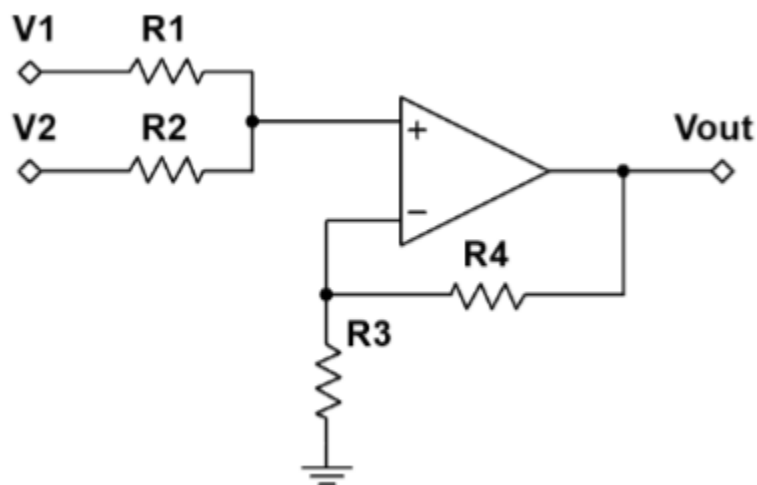
Introduction

When electrical signals travel through a circuit they often pick up unwanted interference called noise. This noise can distort the output signal. Differential signaling is a technique used to compensate for and thus reduce this problem. In differential signaling the same signal is sent in two forms: one is inverted + noise, and the other is normal + noise. Both forms tend to be affected by noise equally, and when a difference amplifier subtracts the two signals, the noise is cancelled out and essentially removed, leaving the original signal.

In this lab noise is added to both a sine wave and its inverted version using PSpice. These two noisy signals are then subtracted using a difference amplifier. The result displayed on the scope simulation is the original sine wave with the noise removed. This lab served as a demonstration of how differential signaling helps minimize noise in electronic systems, as well as a great resource for a review about PSpice setup.

Part 1: Design Opamp Adding Circuit

1. Using opamp adder, build a circuit to add the noise and the sine wave together. Verify the result by performing a time domain transient analysis and recording the waveform for 3 cycles of the input sine wave.



IMG 1. *Non-inverting summer amplifier*

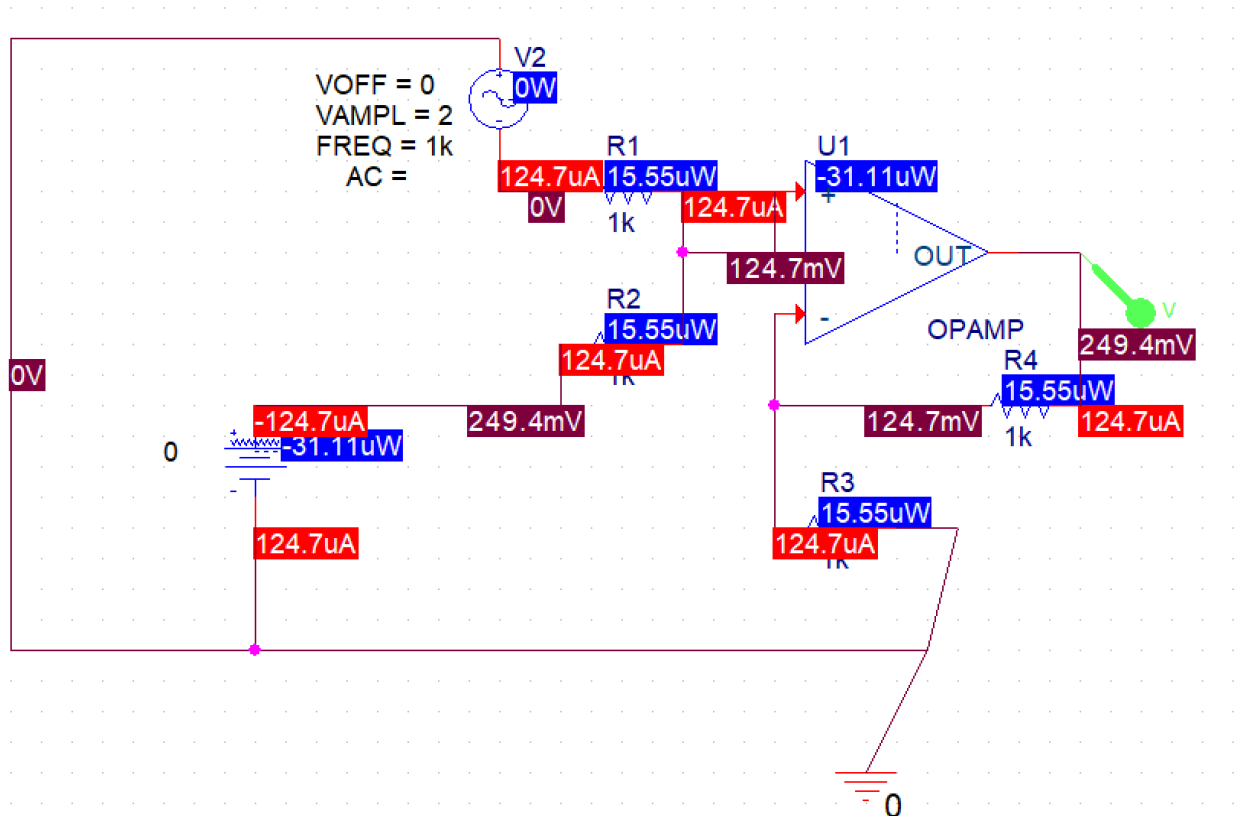
$$V_{OUT} = \left[1 + \frac{R_4}{R_3}\right] \left[\frac{R_2}{R_1 + R_2} V_1 + \frac{R_1}{R_1 + R_2} V_2 \right]$$

$$\text{If } R_1 = R_2 = R_3 = R_4$$

$$V_{OUT} = [1 + 1] \left[\frac{1}{2} V_1 + \frac{1}{2} V_2 \right]$$

$$V_{OUT} = V_1 + V_2$$

IMG 2. Non-inverting summer amplifier calculation of V_{out} , from Professor Mandala's lab handout.



IMG 3. Non-inverting summer amplifier, built on PSpice

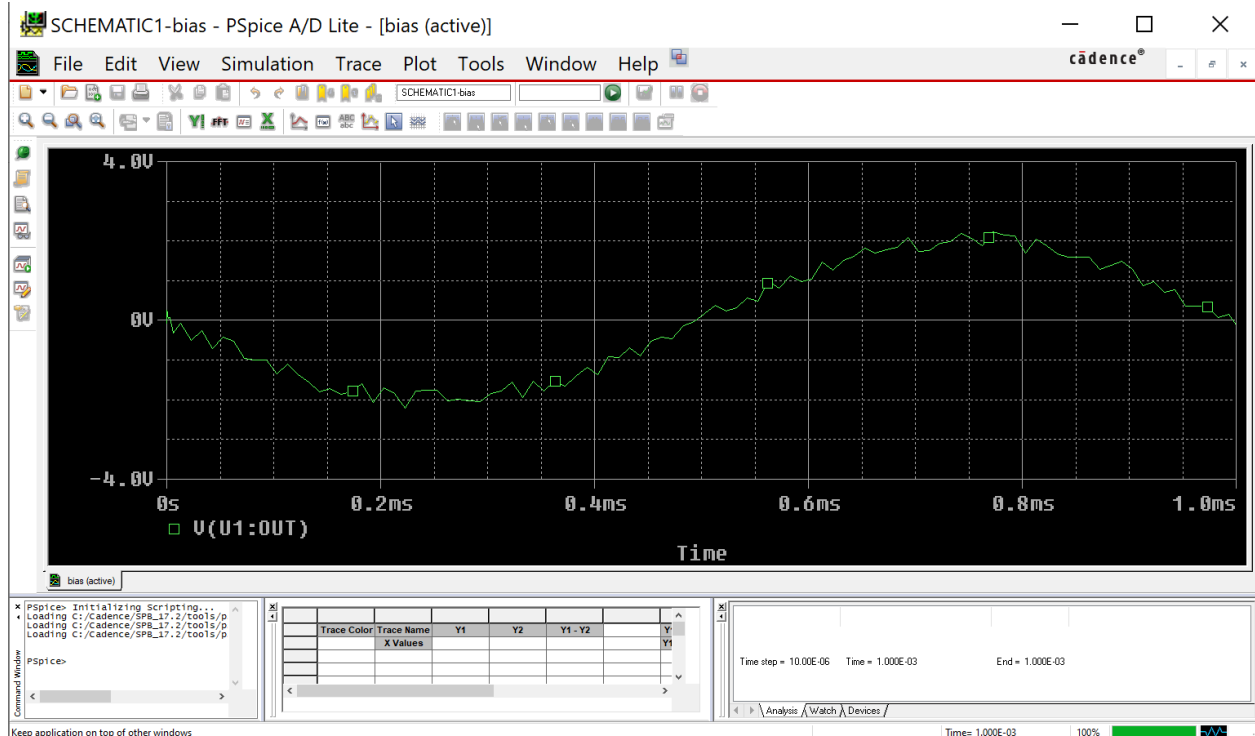
According to the formula for non-inverting summer amplifier in image 2,

$$V_{out} = V_1 + V_2$$

$$V_{out} = 0V + 249.4mV$$

$$V_{out} = 249.4mV$$

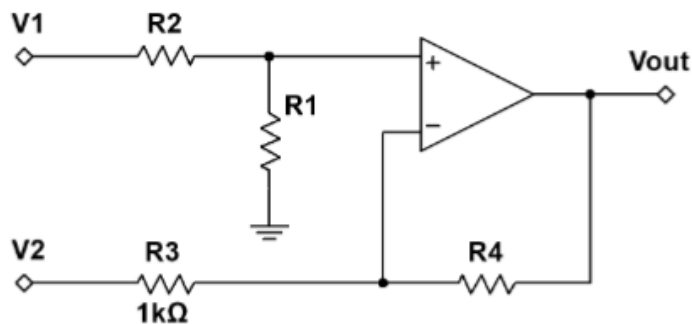
The formula confirms the V_{out} of $249.4mV$, seen in image 3.



IMG 4. Non-inverting summer amplifier scope output from PSpice. Since the sine wave frequency is 1kHz, the period is 1/1kHz or 1ms.

Part 2: Design Difference Amplifier

1. Use a standard difference amplifier op-amp configuration to subtract the two “noisy” signals from each other. Specifically, you will be subtracting the (signal + noise) – (inverted signal + noise).



IMG 5. Difference amplifier

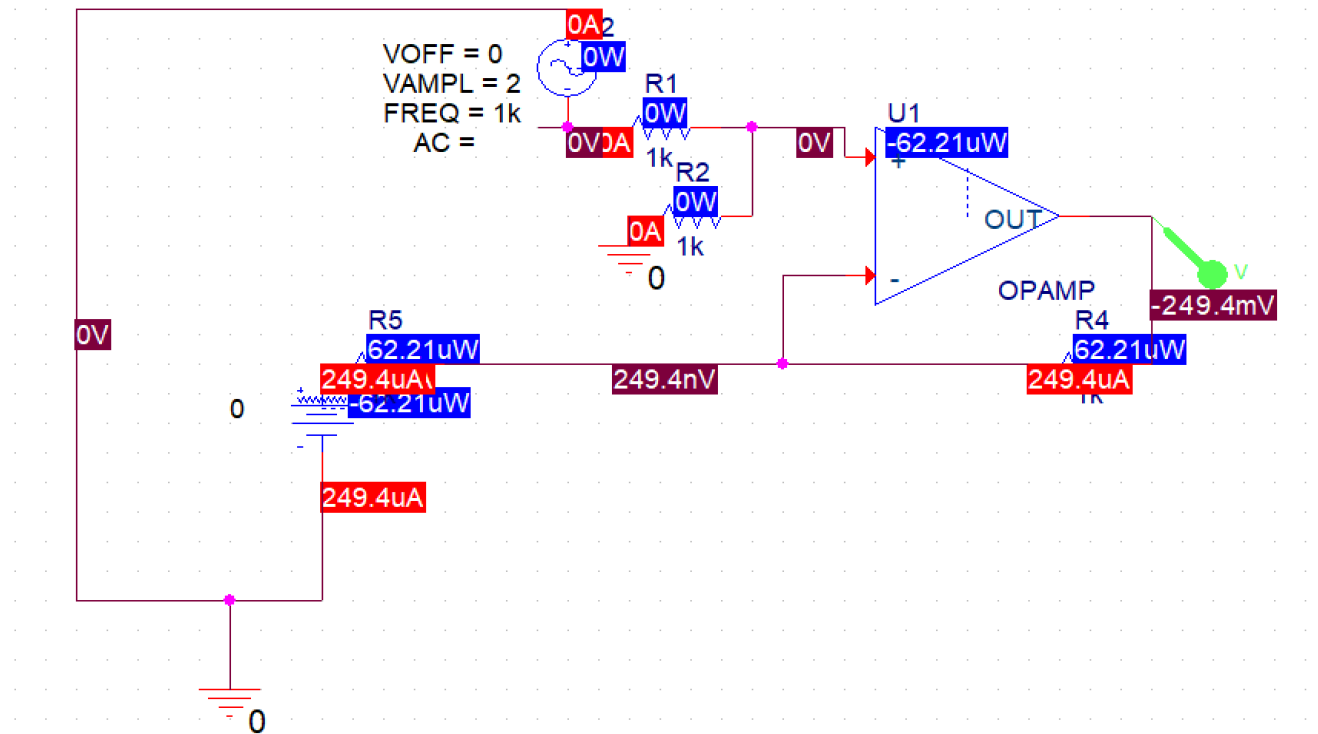
$$V_{OUT} = \left(\frac{R_1}{R_1 + R_2} \right) \left(1 + \frac{R_4}{R_3} \right) V_1 - \frac{R_4}{R_3} V_2$$

$$\text{If } R_1 = R_2 = R_3 = R_4$$

$$V_{OUT} = \left(\frac{1}{2} \right) (1 + 1) V_1 - \frac{1}{1} V_2$$

$$V_{OUT} = V_1 - V_2$$

IMG 6. Difference amplifier calculation of Vout, from Professor Mandala's lab handout.



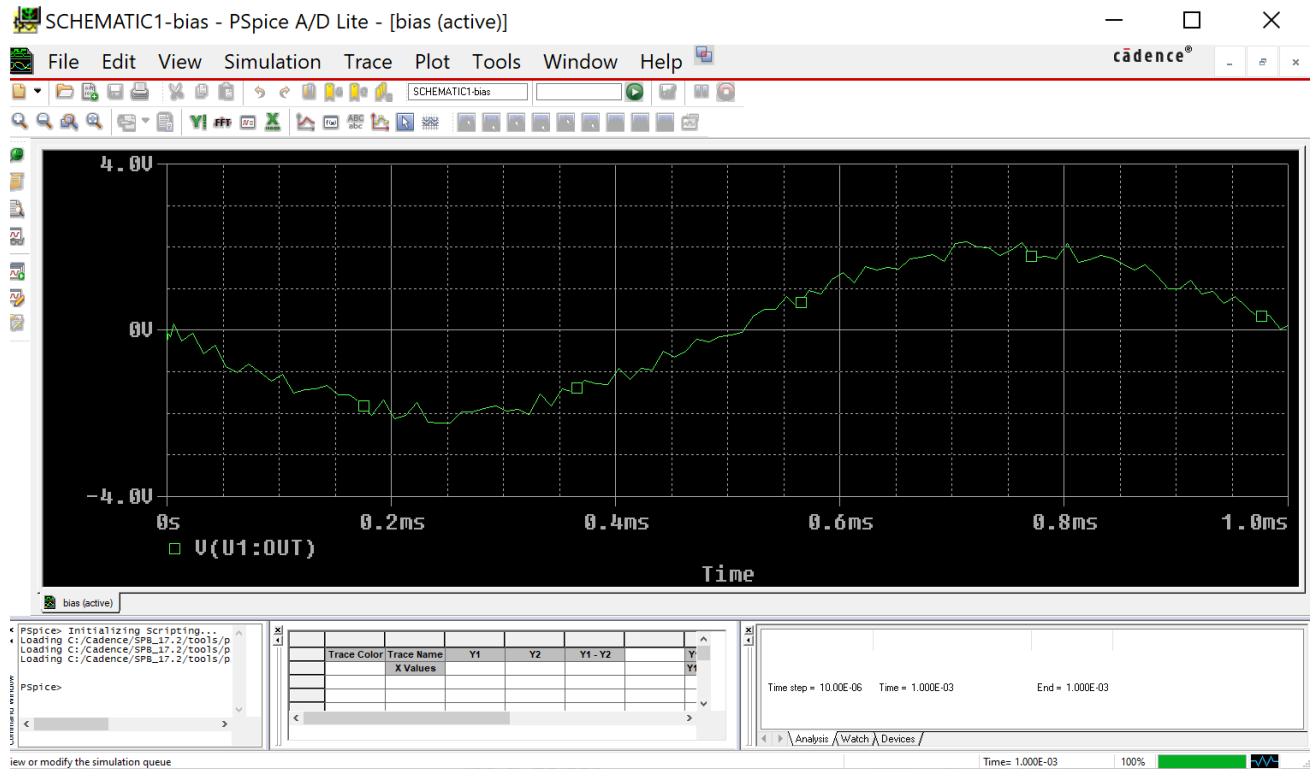
IMG 7. Difference amplifier, build on PSpice

According to the calculations in image 6, $V_{out} = V_1 - V_2$

$$V_{out} = V_1 - V_2$$

$$V_{out} = 0V - 249.4mV$$

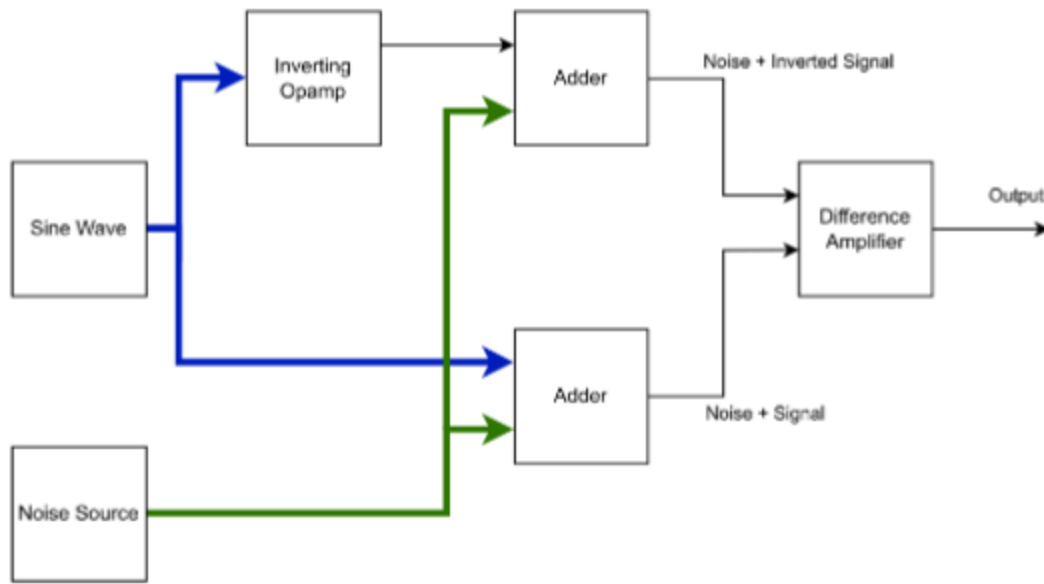
$$V_{out} = -249.4mV$$



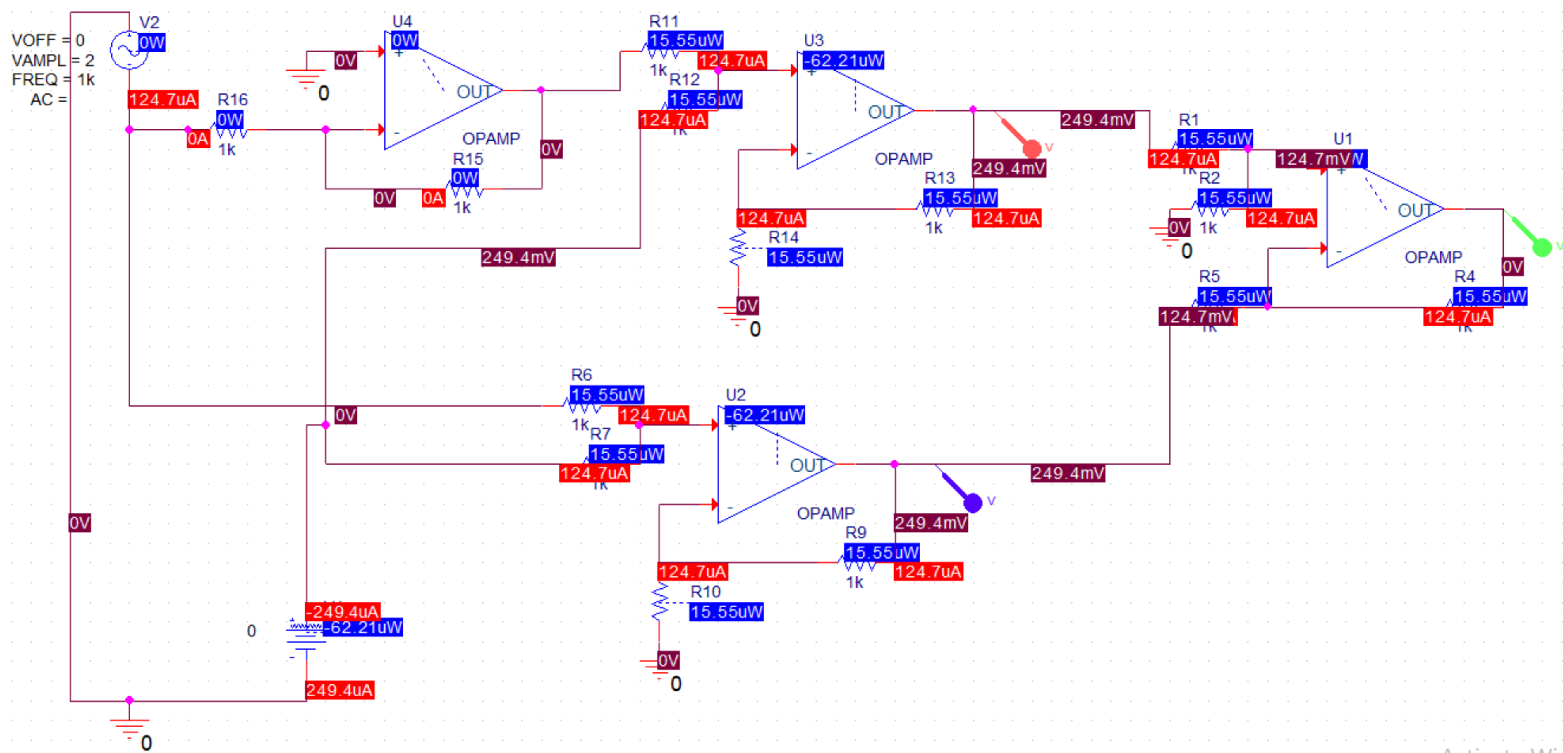
IMG 8. Difference amplifier scope output from PSpice.

Part 3: Recovering the original sine wave

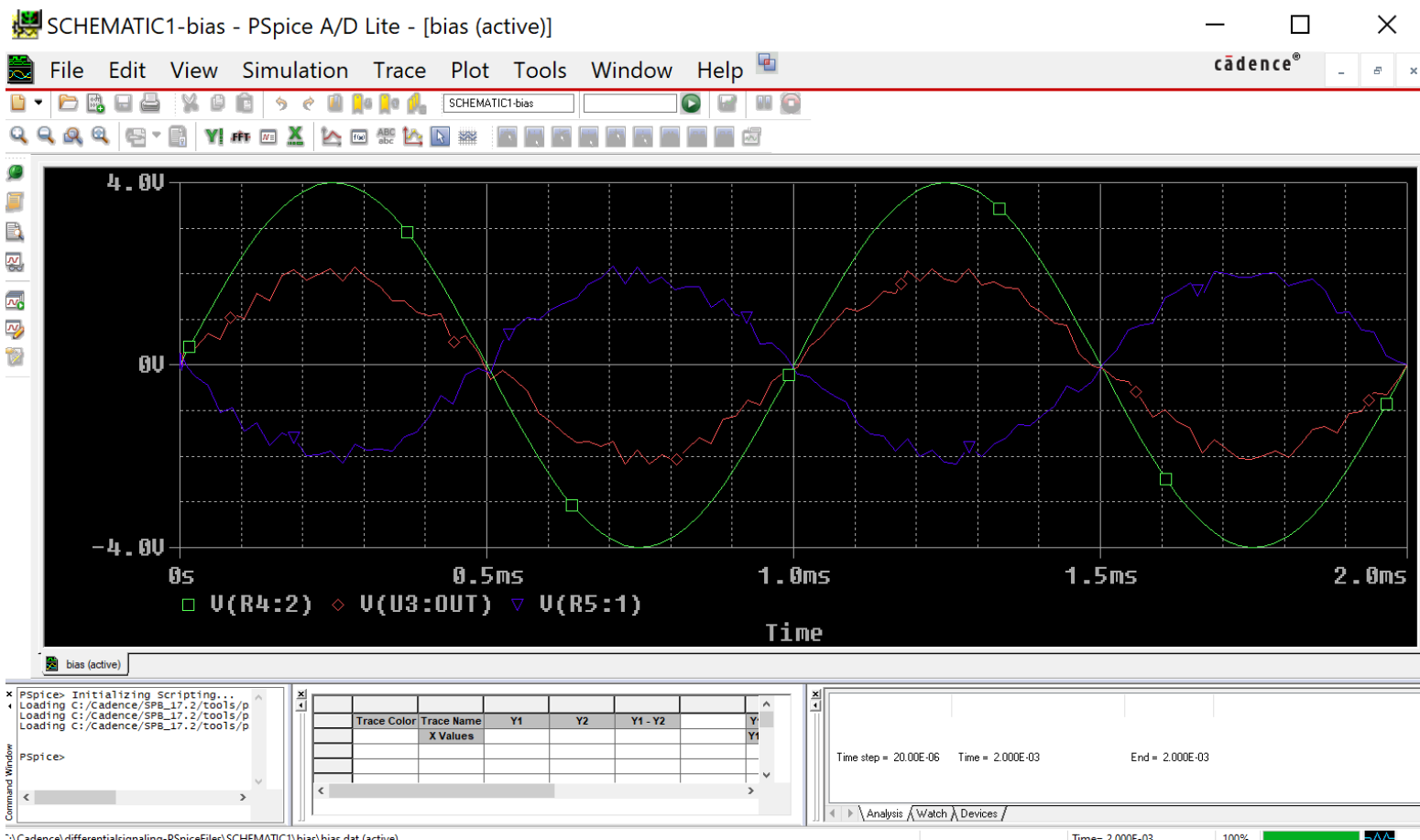
1. The following block diagram in image 9 shows the “big picture” for what we are trying to accomplish.
2. Place probes on: the noisy signal, the noisy inverted signal, and the final output after the difference amplifier. Then perform a time domain transient simulation to show that we are able to recover the original sine wave (with the amplitude doubled) but we have subtracted away the noise. Capture a screenshot to show the noise has been removed.



IMG 9. Overall circuit to build in PSpice



IMG 10. Overall circuit buit on PSpice



IMG 11. Output from PSpice

We can see that there are two noisy signals (blue and red, positive and negative both with noise). The blue and red signals are from when we probe the output of the adders. The difference amplifier subtracts the noise and results in the clean sinewave (green). The amplitude has doubled, which can be seen in image 10 and image 11.

Conclusion

This lab demonstrated how differential signaling can reduce noise in a circuit. By adding the same noise to both a signal and its inverted version and then subtracting them using a difference amplifier, the noise was cancelled.